Week 1

Data Curation and Preprocessing

1 Environment Setup

The notebook begins by installing and importing the libraries needed for the data-preparation stage: `pandas` for tabular data, `datasets` for streaming the parallel corpus, `requests` for downloading files at a fixed revision, and the Indic NLP Library for tokenization. The `transformers` library is installed as well so that it is ready for Week 2, but it is not used in this notebook and no model is loaded. The library versions are printed so that the run can be reproduced: pandas 2.3.3, datasets 5.1.0, transformers 5.19.0 and indic-nlp-library 0.92. An output folder, `data/processed`, is also created for the files saved at the end.

2 Load IndicSentiment

The labelled sentiment data comes from the AI4Bharat IndicSentiment dataset, which contains product and service reviews that have been translated into several Indic languages. The notebook uses Hindi (hi), Bengali (bn) and Marathi (mr). The raw JSON files are downloaded directly from the dataset repository at a pinned revision, so the same data is obtained every time the notebook is run:

`dc8f3f66886531c6897fedffca1e938a68fc5013`

Only a validation split and a test split exist for this dataset; there is no training split. Each language contributes 156 validation reviews and 1,000 test reviews, giving

$$N_{\text{sent}} = 3 \times (156 + 1000) = 3468$$

rows. An assertion confirms that the columns `INDIC REVIEW` and `LABEL` are present before they are copied to `text_raw` and `label`.

3 Load Samanantar Sample

Samanantar is a large English–Indic parallel corpus. Each entry pairs an English sentence (`src`) with its translation in an Indic language (`tgt`). The full corpus is not needed at this stage, so it is streamed and only the first 5,000 sentence pairs are kept for each of Hindi, Bengali and Marathi, giving $3 \times 5000 = 15{,}000$ pairs. Because these are the first rows of the stream rather than a random sample, they should not be treated as representative of the whole corpus.

4 Dataset Exploration

Both datasets are inspected before any processing. The sentiment table has 14 columns, including review metadata (category, product, brand, aspects), the English review, the label and the Indic review. The label column contains 1,743 Positive reviews, 1,719 Negative reviews and

6 rows with no label. No Neutral label occurs anywhere in the data, even though the dataset card describes positive, negative and neutral sentiment. The notebook also checks for an intent column and finds none. The review text itself is never missing; the only missing values are in `LABEL` (6), `BRAND` (24) and `ASPECT COMBO` (21).

The first rows of the three languages show the same review (for example, the same soundbar review) in Hindi, Bengali and Marathi, and the sequence of labels is identical across the three languages in both splits†. The sentiment data therefore appears to contain translations of the same reviews rather than independent samples, which matters for how it can later be split. The parallel sample contains 5,000 pairs for each language.

5 Text Cleaning

A single function, `clean_text`, is applied to all text. It is deliberately conservative: it removes material that carries no sentiment or intent and leaves everything else unchanged. In order, it applies Unicode NFC normalization; removes URLs, e-mail addresses, `@`mentions and HTML-like tags; keeps the words of hashtags (the `#` sign is dropped and underscores become spaces); removes invisible formatting characters (zero-width space, left-to-right and right-to-left marks, soft hyphen and byte-order mark); converts digits from any script to ASCII digits; limits exaggerated repetition; and collapses extra whitespace. Repetition is limited by the rule

$$x^k \longrightarrow x^3 \quad (k \geq 4),$$

where x is any character that is neither a space nor a digit, so that "`!!!!!!`" becomes "`!!!`" but the emphasis is not lost completely. Emojis, punctuation and letter case are left untouched, because they can carry sentiment. The zero-width joiner and non-joiner are also preserved, since they change how conjunct characters are formed in Indic scripts.

6 Testing the Cleaning Function

Before the function is applied to the data, it is tested on four short examples. A Hindi sentence containing a URL, a mention, an emoji and a hashtag keeps its words, its emoji and the hashtag text, and loses the URL, the mention and the hashtag sign. A Bengali sentence with six exclamation marks, an e-mail address and Bengali-script digits ends with three exclamation marks, no e-mail address and the digits written as `123`. A Marathi sentence with a `www` address, an emoji and a hashtag loses the address and the hashtag sign and keeps the rest. Finally, a string containing both the zero-width joiner and the zero-width non-joiner passes through unchanged, and an assertion in the notebook confirms that both characters survive.

7 Applying the Cleaning

The function is applied to the sentiment reviews and to both sides of the parallel pairs, producing the columns `text`, `english_text` and `indic_text`. Cleaning changed 1,204 of the 3,468 sentiment reviews (34.7%) and 3,324 of the 15,000 Indic sentences (22.2%). Most of these changes are invisible: Unicode normalization altered the code points in 1,095 of the changed reviews and 2,791 of the changed Indic sentences†, without changing the characters a reader sees. Digit conversion changed 831 Indic sentences but only one review†.

Two observations limit what these datasets can show about the cleaning step. First, neither dataset contains any emoji†, so the emoji decision has only been exercised on the test strings. Second, URLs and hashtags are rare (4 and 7 Indic sentences respectively, and none of the

reviews†), which is expected for translated reviews and formal text; the datasets are not social-media text. The number of zero-width joiners and non-joiners is unchanged by cleaning, at 51 in the reviews and 958 in the Indic sentences†.

8 Tokenization

The cleaned text is tokenized with the rule-based tokenizer of the Indic NLP Library, called with the language code of each text. Unlike splitting on spaces, it separates punctuation from words, including the danda sentence terminator, commas, question marks and hyphens, so that a word is not stored together with the mark attached to it.

9 Data Quality Checks

Several checks are run on the sentiment data. There is no missing raw text, no empty text after cleaning, and no label outside the expected set (positive, negative, neutral, compared after lower-casing). Six labels are missing, two per language, all in the test split†. There are 6 duplicated texts within the same language and split, which corresponds to two repeated reviews in each language†.

10 Split Overlap and Parallel-Pair Checks

No review text appears in both the validation and the test split. For the Samanantar sample there are no empty English sentences, no empty Indic sentences and no duplicate pairs. The script check flags 42 of the 15,000 Indic sentences (0.28%): 24 in Hindi, 17 in Bengali and 1 in Marathi†. The six flagged sentences that were inspected contain embedded English phrases, headlines or numbers, so they look like mixed-language text rather than extraction errors. These checks confirm that the pairs are present and distinct; they do not verify that each Indic sentence is a correct translation of its English partner. The label counts are identical in the three languages: in the test split, 492 Negative and 506 Positive reviews each, and in the validation split, 81 Negative and 75 Positive reviews each.

11 Saving the Processed Data

The processed data is written to two UTF-8 CSV files in `data/processed`:

- `indic_sentiment_processed.csv` has 3,468 rows and the columns `language`, `split`, `text_raw`, `text`, `label` and `tokens`. The token list is stored as a JSON string so that Indic characters remain readable.
- `samanantar_parallel_sample.csv` has 15,000 rows and the columns `language`, `english_raw`, `indic_raw`, `english_text` and `indic_text`.

Both files read back with these row counts†. They are listed in `.gitignore` and are not committed to the repository, because of the dataset licences; running the notebook recreates them.

12 Findings and Next Steps

The pipeline runs from download to saved files, and cleaning and tokenization preserve the Indic characters and joiners. The checks found no missing review text, no empty text after

cleaning, no label outside the expected set, no overlap between the validation and test splits, and no empty or duplicate sentence pairs. The main limitations, and the work they imply, are as follows.

- **Two sentiment classes only.** The data has Positive and Negative labels but no Neutral, so the three-class sentiment head planned for Week 2 needs another source or a change of design.
- **No intent labels.** None of the datasets used has an intent annotation, so a separate source for intent is still needed.
- **No training split.** IndicSentiment has only validation and test data, and its three languages appear to contain translations of the same reviews. A training split should therefore be made per review, not per row, so that a review is never in training in one language and in testing in another. This needs an explicit key linking the translations; the saved file has none, because rows are aligned only by position.

Subword tokenizers and a pretrained multilingual encoder are planned for Week 2; no model has been loaded or evaluated yet.